Chromosomes I

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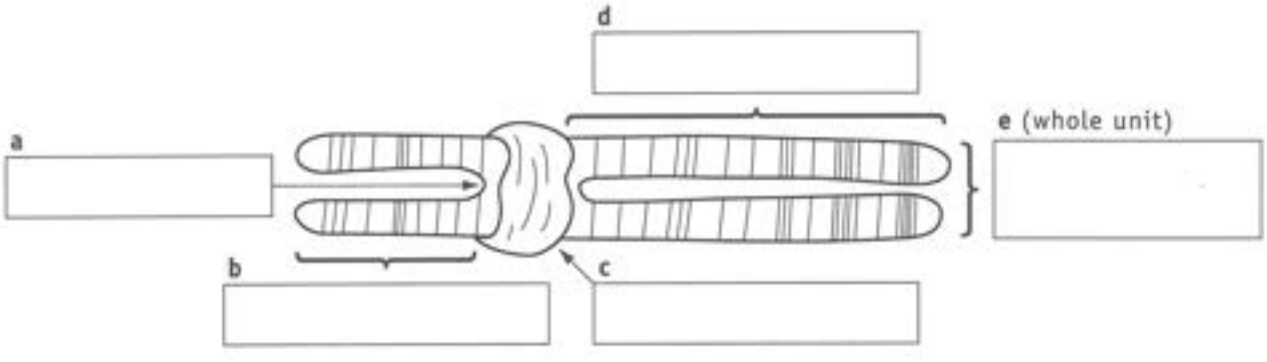
DNA: various nucleic acids that are the basis of heredity and are localised in the nuclei of cells.

Genetic material is found in the form of chromosomes, made of DNA (**d**eoxyribo**n**ucleic **a**cid) and proteins. DNA has a backbone with a tightly-coiled, long double-helix structure (resembling a spiral staircase). The material of chromosomes is called 'chromatin', which means 'coloured threads'. This is what genetic material resembles after it has been stained and viewed under a microscope.

Chromatids: pairs of parallel strands of a duplicated chromosome joined by a single centromere.

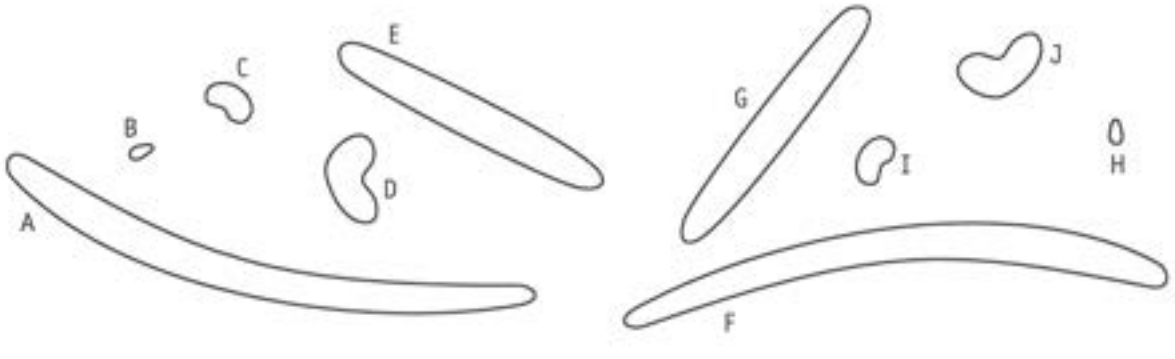
When genetic material replicates, the two copies remain attached for a time. While in this state, each thread-like structure is called a *chromatid*. The pair are referred to as *sister chromatids*. This difference in naming tells us that the genetic material is still undergoing replication. The chromatids are attached at one area called the *centromere*. A protein 'belt' (a *kinetochore*) forms around the centromere. Each part of the sister chromatids is referred to as an arm. The short arms are called *p*-arms (from the French, *petit* meaning 'small') and the long arms are *q*-arms.

1 Reread the text above and label the diagram below.



Chromosomes tend to stay separate most of the time but, during meiosis, this changes. In humans each of the 23 chromosomes contributed by the mother has a matching chromosome contributed by the father, giving a total of 23 pairs. These matching pairs are called *homologous pairs*, or *homologues*.

2 Use five different colours (one per pair) to show which of the chromosomes in the diagram below belong together.



3 What is the shape of a DNA molecule?

4 What is a 'chromatid'?

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